Multivariate Data Analysis Project

Data Analysis

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Introduction

In this project, we apply dimensionality reduction and classification techniques from the second part of the data analysis course to explore and interpret a Spotify dataset, which consists of 1,686 high-popularity tracks and 29 variables, including numerical features (e.g., energy, tempo, danceability, valence) and categorical metadata such as playlist_genre and track_artist. The goal is to investigate relationships between audio features and musical genres.

The preprocessing begins with inspecting dimensions, variable names, and missing values—only one row with a missing track_album_name is removed. Irrelevant metadata (IDs, URLs, musical keys) is excluded, and numeric conversions ensure consistent data types. To narrow the scope, we retain the top 5 most frequent genres and their top 5 subgenres, resulting in the sp_filtered dataset, optimized for multivariate analysis.

To extract meaningful structure from the Spotify dataset, we apply several dimensionality reduction and classification techniques. Each method serves a distinct purpose and provides complementary insights. We begin with **Principal Component Analysis (PCA)**, a technique designed to reduce dimensionality by identifying new uncorrelated variables—called principal components—that summarize the variance present in a set of potentially correlated features. These components are linear combinations of the original variables and are derived from the covariance matrix using eigenvalues and eigenvectors. PCA allows us to retain most of the variability in the data while representing it in a lower-dimensional space, making it suitable for visualization (e.g., biplots) and for detecting linear relationships between variables.

Next, we use **Multidimensional Scaling (MDS)**, which projects observations from a high-dimensional space into a lower one while preserving the pairwise distances between them. Unlike PCA, MDS does not assume linearity. We apply three variants: metric MDS with Euclidean distance, metric MDS with Gower distance (which handles mixed data types), and non-metric MDS based on rank-order dissimilarities. The goal is to obtain a configuration of points where distances in low dimensions reflect real-world similarities, facilitating the interpretation of clusters or underlying dimensions.

To handle exclusively categorical data, we implement **Multiple Correspondence Analysis (MCA)**. This method transforms a set of categorical variables into a geometric representation in reduced dimensions, allowing us to analyze associations between categories and visualize patterns. MCA can be computed from the indicator matrix (preserving information on individuals) or the Burt matrix (contingency table of all possible variable combinations). The positions of the categories in the resulting space reveal their relationships, while proximity to the origin indicates balance across levels.

Cluster Analysis is an unsupervised learning technique that seeks to group similar observations into distinct clusters based on their multivariate characteristics. The main objective is to uncover hidden structures in the data by maximizing homogeneity within groups while ensuring heterogeneity between them. This is achieved by measuring distances or dissimilarities between observations and applying algorithms, such as hierarchical or partitioning methods, that organize the data into meaningful subgroups. Cluster analysis is particularly useful for discovering patterns, identifying natural groupings, and supporting data-driven segmentation without relying on predefined categories.

Lastly, we use **Quadratic Discriminant Analysis (QDA)** and **Multivariate Analysis of Variance (MANOVA)**. QDA is a supervised classification technique that models each group with its covariance matrix, allowing flexible, non-linear decision boundaries. It is particularly useful when class variances differ across categories. MANOVA, on the other hand, assesses whether the multivariate means of the groups differ significantly, accounting for correlations among the dependent variables. Together, these methods help validate the separation observed in previous analyses and assess the predictive potential of the musical features with genre classification.

Principal Component Analysis (PCA)

We began the analysis by selecting only the numerical variables, while treating the categorical ones as supplementary in order to enhance the interpretability of the results. After preparing the dataset, we applied the PCA() function to extract the principal components and their associated eigenvalues.

To determine the appropriate number of dimensions to retain, we considered several established criteria, including the **Kaiser rule** (retaining components with eigenvalues greater than 1), the **cumulative explained variance** criterion (retaining enough dimensions to explain at least 80% of the variance), and the **Scree plot** method (identifying the "elbow" in the eigenvalue curve). After comparing these options, we opted for the elbow criterion in the Scree plot.

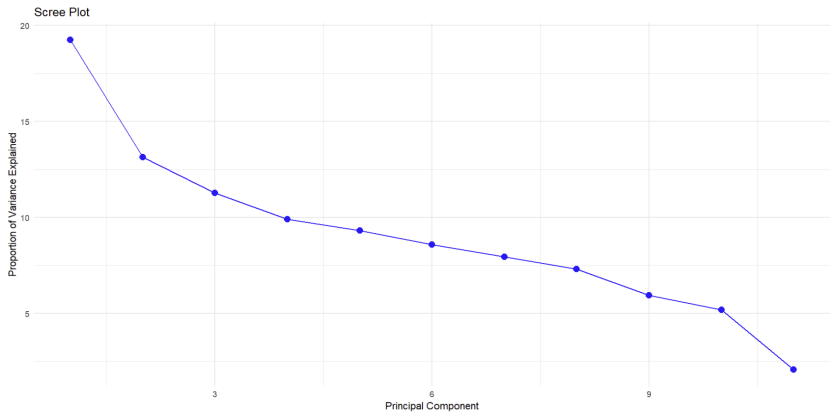

Figure 1: Scree plot representative of the proportion of variance explained depending on principal components

As observed in Figure 1, the elbow in the eigenvalue curve appears at the fourth component. Consequently, we retained the first four dimensions as the principal components for further analysis. We computed the Kaiser-Meyer-Olkin (KMO) measure to assess the sampling adequacy for factor analysis. However, as most variables yielded MSA values below the commonly accepted threshold of 0.5, we decided not to rely on this test and proceeded with dimensionality reduction techniques independently. Once the dimensionality was reduced, we proceeded to interpret the underlying meaning of each component through both analytical and exploratory approaches.

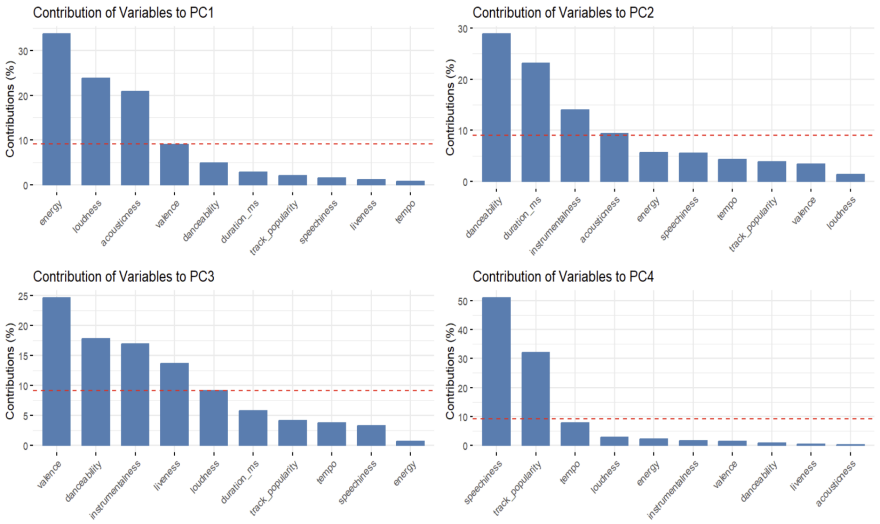

Figure 2: Scree plot representative of the proportion of variance explained depending on principal components

	Dim.1	***Dim.2***	***Dim.3***	***Dim.4***
energy	*0.84466418*	*-0.28511238*	*-0.09014220*	*-0.15074185*

tempo	*0.11886662*	*-0.25006443*	*-0.21412811*	*0.29114468*
danceability	*0.31959463*	*0.64577826*	*0.46928017*	*0.09105391*
loudness	*0.71013285*	*0.13815524*	*-0.33572138*	*-0.17448665*
liveness	*0.15006580*	*-0.09771314*	*-0.41154564*	*0.06610065*
valence	*0.43487053*	*0.22145950*	*0.55300552*	*-0.11892513*
speechiness	*0.17773455*	*0.28300910*	*-0.20124986*	*0.74645841*
track_popularity	*-0.20425219*	*0.23502673*	*-0.22539275*	*-0.59098322*
instrumentalness	*0.02813248*	*-0.44951097*	*0.45795659*	*0.13226035*
duration_ms	*-0.24023021*	*-0.57802085*	*0.26630668*	*0.00052050*
acousticness	*-0.66368483*	*0.36779220*	*-0.07852612*	*0.04166440*

Table 1. *Factor loadings of the variables on the five extracted dimensions*

The interpretation of the dimensions was guided by the factor loadings, which revealed distinct patterns across the four extracted components.
Dimension 1 is primarily defined by energy (0.84), loudness (0.71), and valence (0.43), with a strong negative contribution from acousticness (-0.66). This dimension separates high-energy, loud, and positive tracks from softer, acoustic, and mellow ones. The moderate contribution of danceability (0.32) reinforces its interpretation as an axis of commercial appeal. **Dimension 2** is characterized by positive loadings from danceability (0.64) and negative contributions from instrumentalness (-0.44) and duration_ms (-0.58). It reflects a contrast between short, danceable tracks and longer, more instrumental pieces, capturing an axis of rhythmic accessibility versus musical complexity. **Dimension 3** shows positive correlations with valence (0.55), danceability (0.47), and instrumentalness (0.45), while being negatively associated with liveness (-0.41) and loudness (-0.33). This suggests it differentiates bright, expressive tracks with instrumental richness from louder, performance-driven songs. **Dimension 4** is most influenced by speechiness (0.74), with moderate contributions from tempo (0.29) and a strong negative loading from track_popularity (-0.59). It contrasts spoken, less mainstream tracks with melodic, widely popular songs, capturing a lyrical vs. commercial polarity.

Furthermore, a graphical analysis of the correlation circles was performed to visually assess the relationship between the variables and the extracted dimensions, helping to assign meaningful labels to each.

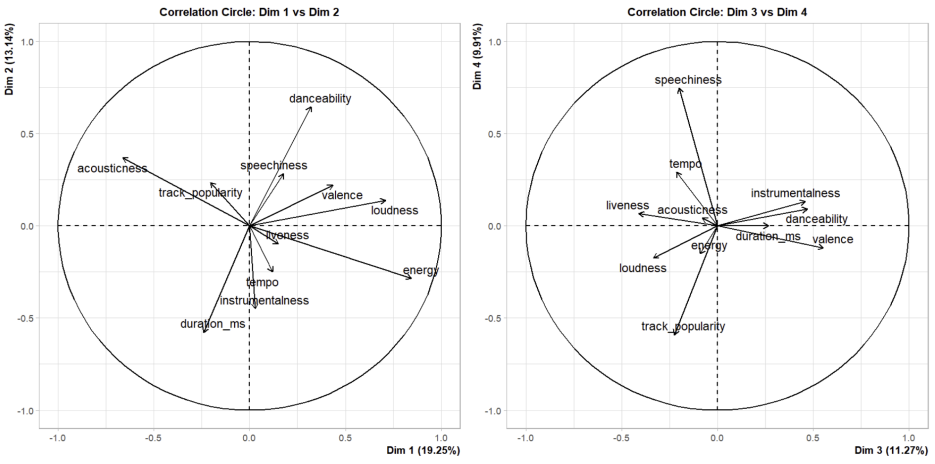

Figure 3: Comparison of Correlation Circles for Principal Component Analysis (PCA)

Dimension 1 vs. Dimension 2
In **Dimension 1**, strong positive correlations appear with *energy*, *loudness*, *valence*, and *danceability*, while *acousticness* and *duration_ms* are negatively correlated. This dimension separates energetic, lively, and positive songs from acoustic and calmer ones, and can therefore be labeled **Commercial Sound and Loudness**. **Dimension 2** distinguishes *danceability* and *acousticness* from *instrumentalness* and *duration_ms*, highlighting a contrast between danceable, concise tracks and longer, instrumental ones. It is thus named **Danceability and Brevity**.

Dimension 3 vs. Dimension 4
Dimension 3 separates *valence*, *danceability*, and *instrumentalness* from *liveness* and *loudness*, suggesting a differentiation between joyful, danceable, and instrumental music versus louder, live-oriented tracks. It is labeled **Musical Complexity**. Finally, **Dimension 4** is positively associated with *speechiness* and *tempo*, and negatively with *track_popularity* and *valence*, reflecting tracks rich in lyrical content and fast tempo, often less mainstream. This dimension is labeled **Lyric-Driven (Rap Focus)**.

Subsequently, we explored the distribution of individual observations. Although we initially attempted to visualize the individuals by song title (Appendix 1), the high number of points rendered the plots unreadable. As a result, we opted to plot the individuals using index labels to improve clarity.

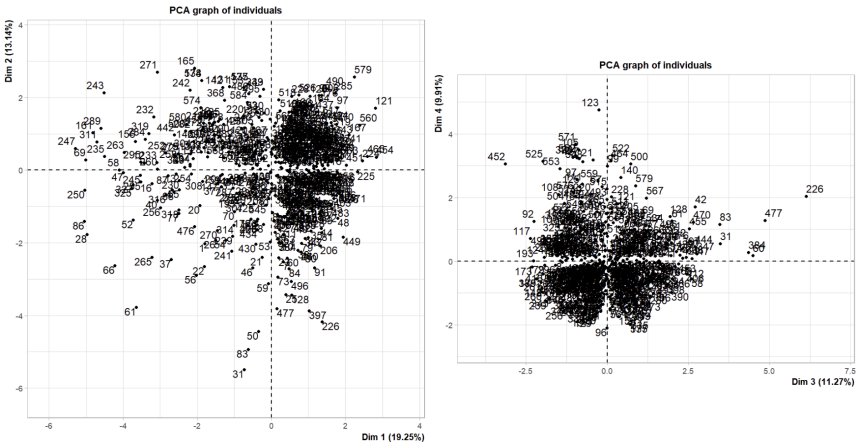

Figure 4: Individual Factor Map: PCA Representation

To complement the interpretation of the correlation circles, we analysed the distribution of individuals on the factorial planes, focusing on those located at the extremes of each dimension to validate our hypotheses. On Dimension 1, *The Largest* (hip-hop) is highly positive, reflecting its loud, energetic, and commercial traits, while *Wish You Were Here* (Pink Floyd) is strongly negative, representing acoustic and emotive qualities. Dimension 2 separates short, danceable tracks like *Dos Días* from longer, complex ones like *Master of Puppets*. Dimension 3 contrasts rich electronic tracks like *Around the World* with simpler songs such as *Thought I Was Dead*. Lastly, Dimension 4 distinguishes lyrical, speech-driven hip-hop (*Blow for Blow*) from melodic mainstream pop (*Timeless*). These examples confirm the proposed interpretations for each dimension. Overall, the analysis of representative tracks at the extremes of each dimension consistently validates the proposed labels and interpretations derived from the variable correlation analysis.

To further support our interpretation of the PCA space, we computed the mean and standard deviation of each genre across the four retained dimensions. This information is presented in **Appendix 2,** together with a visual representation of the individuals colored by musical genre (Appendix 3). These additional analyses confirm the validity of our dimensional interpretations and offer deeper insight into genre-specific patterns in the dataset.

Multidimensional scaling (MDS)

We began by selecting and scaling the numerical variables, followed by the removal of duplicate rows to ensure data integrity. Our initial attempt at metric MDS yielded a very low goodness-of-fit (GOF) value of 30.79%, indicating poor representation. To address this, we examined variable collinearity and eliminated those with high redundancy and lower relevance. We then computed distance matrices using Euclidean distance and reapplied metric MDS via the cmdscale function. This time, the resulting GOF improved substantially to 52.53%. We proceeded to visualize the MDS configuration based on these results.

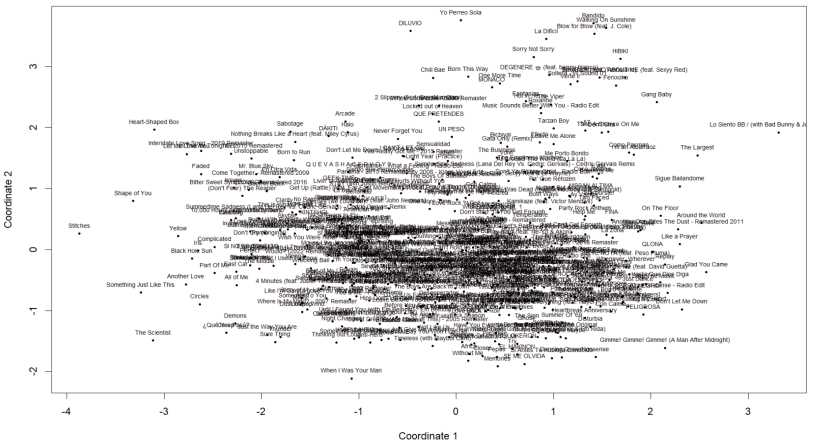

Figure 5: Metric Multidimensional Scaling (MDS) Representation of Songs

The metric MDS projection reveals a high concentration of songs near the center of the plot, likely representing mainstream tracks with average values across most musical variables. Outlier songs, such as *DILUVIO*, located far above the origin, may exhibit unusual characteristics such as extreme tempo or low energy. Similarly, tracks like *Yo Perreo Sola*, *Lo Siento BB:/*, and *HIBIKI* appear distinctly separated from the core cluster, suggesting a strong urban or reggaetón influence. On the opposite side, songs such as *Stitches* and *The Scientist* are positioned further to the left, which may indicate softer, more acoustic, or melancholic features. Songs with moderate energy and instrumentation, such as *Shape of You*, *Yellow*, and *Another Love*, tend to cluster together, whereas high-energy EDM-pop tracks like *On The Floor*, *Around The World*, and *Titanium* are situated toward the upper or right regions of the map. The absence of sharply defined genre-based clusters suggests that the most popular songs exhibit considerable stylistic diversity. Nonetheless, localized groupings may hint at subgenre similarities, though further categorical labeling would be required to confirm these patterns.

To enrich the interpretation, we developed additional visualizations. The first alternative consisted of plotting the individuals colored by musical genre, enabling a clearer examination of genre-specific clusters and stylistic similarities across the MDS space.

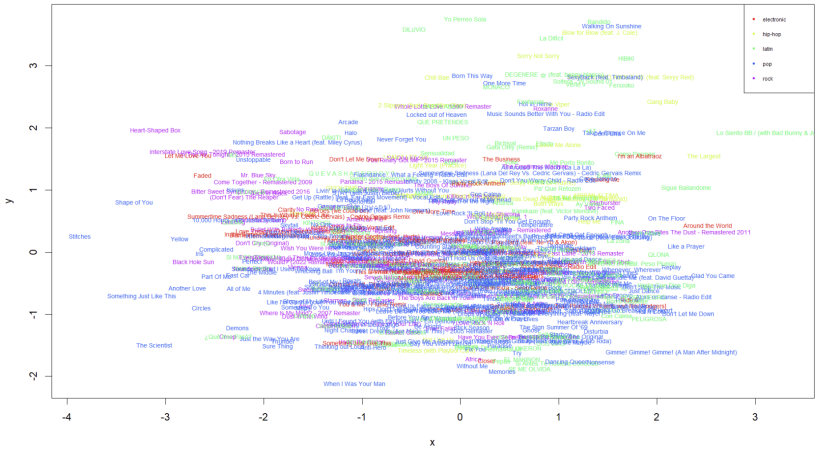

Figure 6: Metric Multidimensional Scaling (MDS) Representation of Songs by Genre

The MDS plot colored by musical genre highlights subtle yet meaningful patterns in the distribution of songs. Most tracks remain concentrated near the origin, suggesting that many share similar acoustic profiles. However, some genre-based tendencies begin to emerge. Latin songs, for instance, appear predominantly in the upper right quadrant, indicating a tendency toward high energy and danceability. Hip-hop tracks tend to occupy the upper central area, possibly reflecting their characteristic high *speechiness* and rhythmic structure. Acoustically similar genres—such as hip hop, rap, and trap—cluster closely, emphasizing their shared musical traits. In contrast, more traditional or stylistically distinct genres like classical, rock, or metal are spatially distant from urban genres such as reggaetón, reflecting their contrasting audio features.

To further support our observations, we included in the **Appendix** a dedicated plot of the 200 most popular tracks. This allowed us to extract additional insights regarding mainstream musical trends and their spatial distribution in the MDS configuration. We also experimented with rotating the MDS configuration in an attempt to enhance interpretability. However, since the rotation did not provide additional insights or improve the clarity of the results, we chose to omit it from the main analysis. Moreover, we visualized the MDS layout based on several individual musical variables. These plots allowed us to examine how specific characteristics—such as energy, acousticness, or valence—are spatially organized across the MDS dimensions.

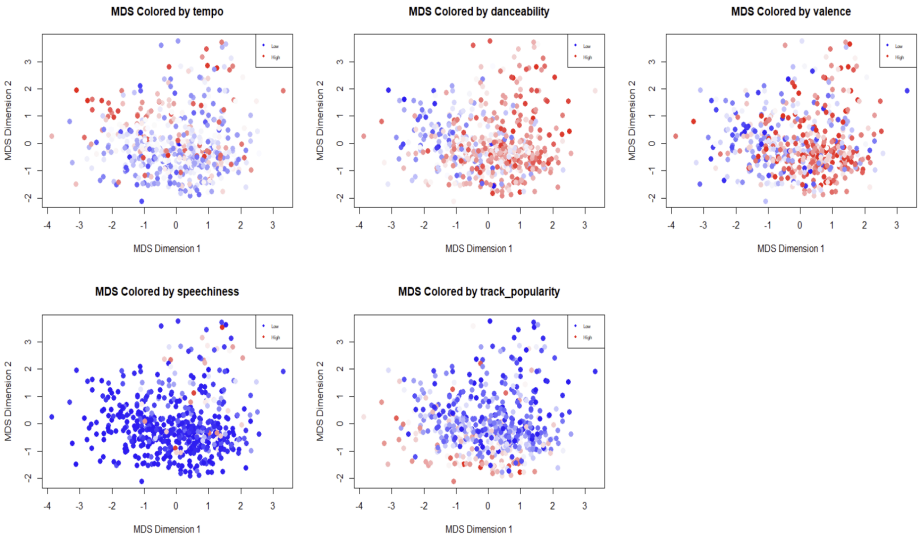

Figure 7: MDS Projection of Songs Colored by Acoustic Features

In the case of *danceability* and *valence*, there is a clear gradient along Dimension 1, with higher values clustering on the right-hand side, suggesting that this dimension captures the upbeat and energetic nature of tracks. Conversely, *track_popularity* and *tempo* show more pronounced variation along Dimension 2. Specifically, higher values of *tempo* tend to concentrate in the upper region, while higher popularity appears toward the bottom, reinforcing the interpretation of Dimension 2 as reflecting a trade-off between lyrical density and mainstream appeal.

Based on both the visual distribution of genres and the correlations with musical features, the two dimensions extracted from the metric MDS can be meaningfully interpreted. **Dimension 1** appears to capture a combination of rhythmic energy and emotional positivity. Its strong positive correlations with danceability and valence suggest that this axis differentiates between tracks that are upbeat and feel-good (such as pop or Latin) versus those that are more subdued or instrumental. This dimension can therefore be interpreted as representing **"Expressive and Rhythmic Intensity"**, where higher values reflect vibrant, danceable, and lyrically rich music, while lower values correspond to calmer, less expressive tracks.

Dimension 2, in contrast, shows a strong positive association with tempo and speechiness, and a moderate negative correlation with track popularity. This suggests a separation between fast-paced, lyrically dense songs—often found in genres like rap or experimental hip-hop—and more melodic, widely consumed tracks. As such, this dimension may reflect a **"Lyrical Density vs. Mainstream Appeal"** axis, with higher values indicating more rhythmically and verbally complex pieces, and lower values aligning with popular, melodic content.

Next, we applied Multidimensional Scaling (MDS) using the **Gower distance**, a dissimilarity measure suitable for datasets containing both **numerical and categorical variables**. Unlike Euclidean distance, Gower allows the integration of mixed data types, making it ideal for our dataset, which includes musical features alongside categorical attributes such as playlist genre. This approach provides a more comprehensive representation of dissimilarities between songs. After computing the Gower dissimilarity matrix, where we included the playlist genre as a factor, we performed metric MDS and visualized the resulting configuration to examine potential patterns and clusters in the musical space.

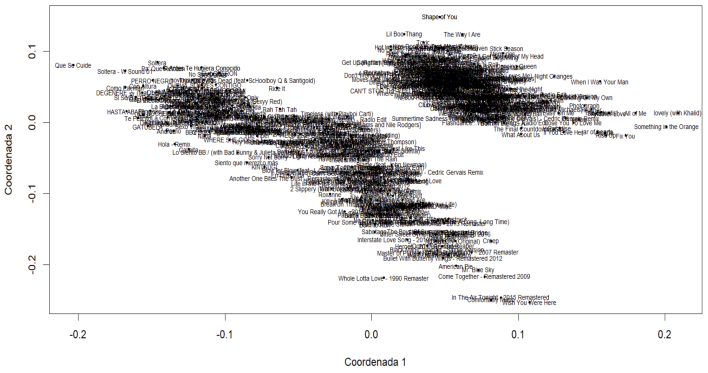

Figure 8: MDS with gower distance

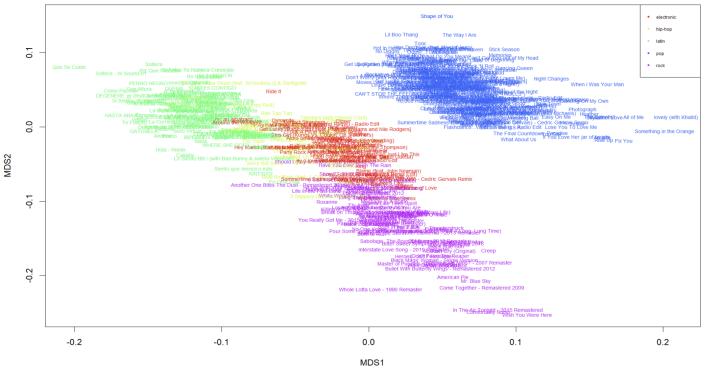

Figure 9: MDS with gower distance by Genre

The MDS plot based on Gower distance provides a notably clearer separation of musical genres compared to the metric version, highlighting the advantage of incorporating categorical variables such as genre. Pop tracks cluster in the upper-right quadrant, while Latin music appears primarily in the upper-left, and rock is situated toward the lower part of the plot. The relative proximity between rock and electronic, or electronic and hip-hop, suggests stylistic or production-based similarities. Let's now visualize the MDS layout based on several individual musical variables.

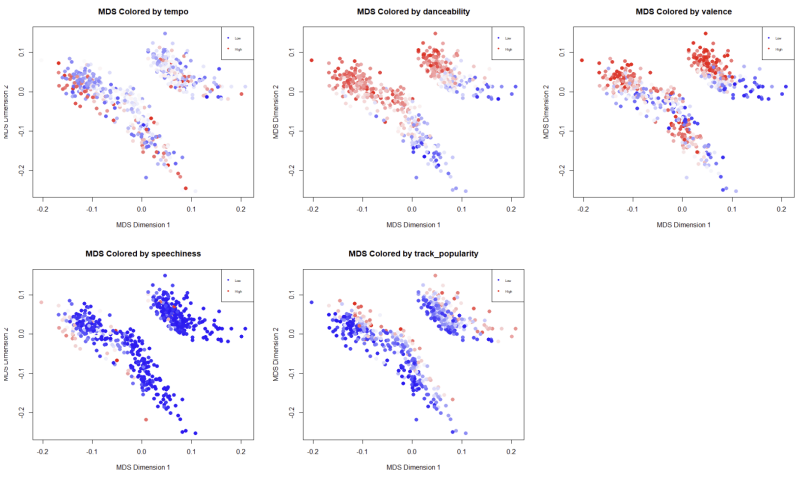

Figure 10: MDS with gower distance Projection of Songs Colored by Acoustic Features

We observe that Dimension 1 is strongly negatively correlated with the variable *speechiness*, suggesting that it may capture the degree of melodic versus spoken content in the tracks. Aside from this, it shows limited association with the remaining variables, indicating a more specific interpretive axis. In contrast, Dimension 2 presents a clearer structure, as it distinctly separates songs with high values of *danceability* and *valence*, highlighting a dimension of emotional positivity and rhythmic appeal. So, the first MDS dimension (horizontal axis) seems to reflect a transition from rhythm-dominated styles to melody-driven structures. Genres such as Latin and hip-hop, located on the left, are typically rhythmically intense and vocally recited. On the right, genres like pop emphasize melodic and harmonic elements. Additionally, tracks on the left tend to exhibit higher speechiness, aligning with genres like hip-hop, confirming the expressive contrast between vocal recitation and traditional singing.

The second dimension (vertical axis) may capture a spectrum of temporal and emotional energy. Tracks positioned at the top—such as modern, danceable pop songs—exhibit high levels of danceability and emotional valence. In contrast, songs toward the bottom, including classic rock and slower, introspective tracks, display lower danceability despite potentially higher tempo.

Overall, the Gower-based MDS reveals coherent genre-based clustering and interpretable latent dimensions that align well with known musical characteristics. In conclusion, the Gower-based MDS outperformed the metric alternative, as it effectively incorporates both numerical and categorical variables, resulting in more meaningful genre-based separations. Although we also explored a non-metric MDS approach, it did not offer any notable improvements or insights beyond those already obtained, and thus, its results are included in the appendix for reference only.

Multivariate Correspondence Analysis (MCA)

The first step in applying Multiple Correspondence Analysis (MCA) is to determine which variables to include in the analysis. Upon reviewing our dataset, we find that although some categorical variables, such as the recording date or the artist's name, are present, they do not appear to be particularly relevant or explanatory for this analysis. Therefore, we opt to categorize numerical variables that have a straightforward range (e.g., from 0 to 1 or 0 to 100), allowing for a more comprehensive examination of our dataset.

Once the variables have been categorized and the indicator matrix has been generated, we proceed with the MCA. However, before delving into the analysis, we must determine the number of dimensions to retain. If we apply the rule of selecting all dimensions with an eigenvalue above the average, we would retain eight dimensions in total. Since this is impractical, we instead identify an "elbow" in the scree plot of eigenvalues to guide our selection of a more manageable number of dimensions

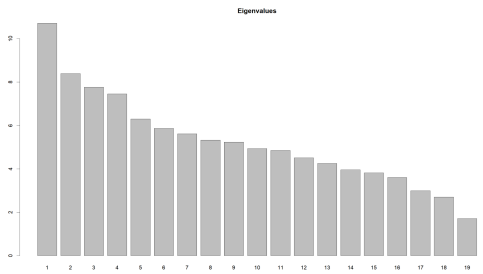

Figure 11: histogram of eigenvalues

By analyzing the graph, we observe a clear elbow at two dimensions, which justifies conducting the analysis using only two. We have assessed the presence of the Guttman effect and found no evidence of it, as can be seen in the plot presented in Appendix 6. The original MCA plot, shown in Appendix 7, does not yield a clearly interpretable representation. Therefore, we approach the analysis through three complementary methods. First, by examining the individual plots colored

by category. Second, through the plot of the categorical variables, and finally, by assessing the relationship between each variable and the two selected dimensions. We begin by analyzing the individual plots based on their categorical grouping.

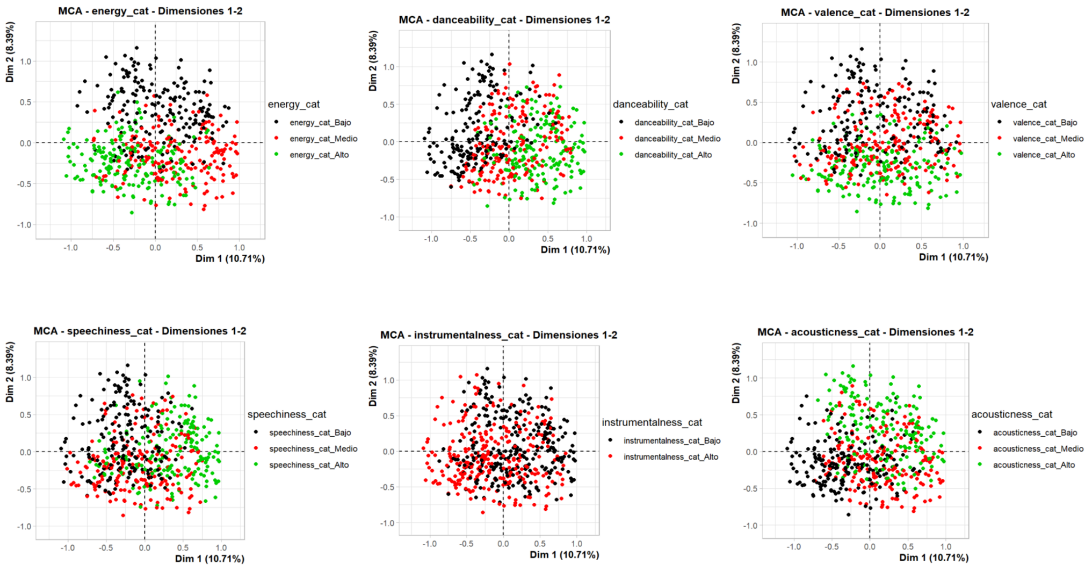

Figure 12: Individual plots by categories

Based on the plots, Dimensions 1 and 2 together help reveal distinct patterns across the analyzed musical attributes. Dimension 1 appears to primarily separate low from higher values in danceability—tracks with higher danceability tend to be positioned more positively along this dimension. For energy specifically, while Dimension 1 separates low-energy tracks from the rest, Dimension 2 further distinguishes between medium and high energy levels. Similarly, speechiness and instrumentalness show variation along Dimension 1, with higher values leaning toward the right and left ends, respectively, though the latter is less pronounced. Dimension 2 plays a more important role in distinguishing valence, as lower values on this dimension correspond to lower valence. Acousticness displays a diagonal trend across both dimensions, where higher values are concentrated in the upper-right and lower values in the bottom-left. Since *liveness* and *popularity* are not clearly represented in the first two dimensions, and their variation appears to be captured in higher components, we decided not to include their plots.

We now turn to the analysis of categorical variables, using visualizations to explore how categories such as genre or subgenre are distributed across the reduced dimensional space. This helps reveal underlying patterns and relationships between groups.

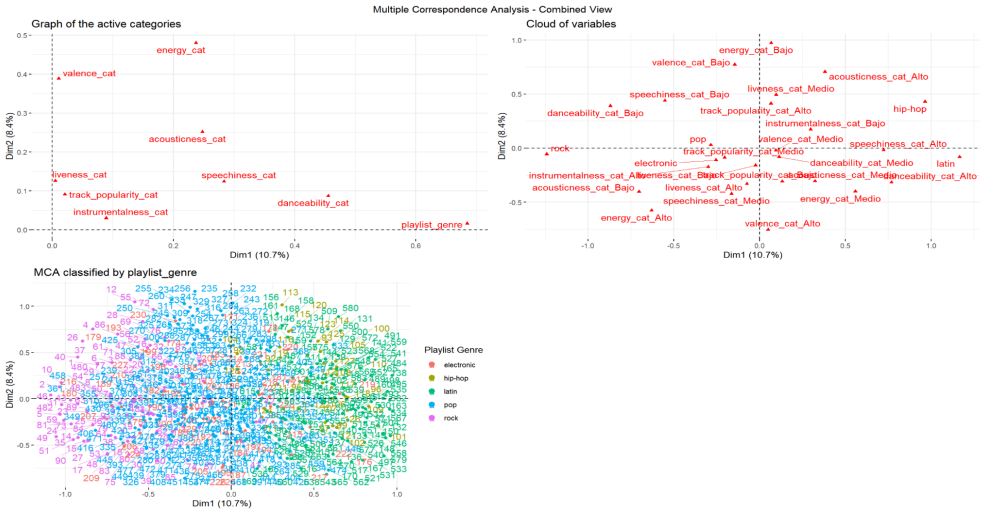

Figure 13: Individual and variable plots

This category-based plot reinforces and clarifies much of what we had already observed earlier. When analyzing the musical genres, we see a clear separation along Dimension 1: genres like hip-hop and Latin cluster on the right side of the graph, while rock is positioned on the left. Pop and electronic lie more centrally, suggesting they are less polarized and perhaps more adaptable across different musical characteristics.

Looking at the graph of the cloud of variables, energy appears to be strongly influenced by Dimension 2, though Dimension 1 also has some influence. Instrumentalness, on the other hand, shows a clearer separation along Dimension 1. Interestingly, while the individual plots did not reveal a strong pattern for danceability, this category-based analysis shows it varies along both dimensions—higher danceability tends to be found in the lower-right area of the graph.

Examining the origin point, we observe that most observations belong to pop and electronic genres, which represent the most generic categories in terms of musical specificity. We can conclude that proximity to the origin, despite containing numerous observations, indicates belonging to generic and commercial music types without distinctive or particularly relevant characteristics.

Finally, the graph of active categories highlights that energy is strongly aligned with Dimension 2, as previously noted, while genre seems more associated with Dimension 1. Interpreting the genres from this perspective, Dimension 1 clearly distinguishes between Latin and hip-hop—genres characterized by higher speechiness (such as in trap or rap) and higher danceability—and rock, which is more instrumentally rich. Pop and electronic genres appear near the center, indicating that they are less extreme in the variables analyzed and may blend features from both sides of the spectrum.

However, Dimension 2 presents greater interpretive challenges when examining the individual genre plots. Nevertheless, by analyzing other visualizations, particularly both categorical variable plots, we can observe that this dimension exhibits a strong relationship with energy and valence. This association provides valuable insight into the difficulty of interpreting Dimension 2 at the genre level, as the vast majority of musical genres encompass songs with considerable diversity in these emotional and energetic characteristics. This heterogeneity within genres suggests that energy and valence operate as cross-cutting dimensions that transcend traditional genre boundaries, reflecting the complex and multifaceted nature of contemporary musical expression.

Cluster Analysis

The first step before performing the Cluster Analysis was to scale the data and compute the distance matrix. Once this was completed, we proceeded to evaluate several hierarchical clustering methods to determine the most suitable approach. The methods tested included: Single linkage, Complete linkage, Average linkage, Ward's method, and Centroid method; we can see all of them at the Appendix. After analyzing the resulting dendrograms, we observed that, despite some interpretative challenges, the most balanced tree structure was produced by Ward's method. By "balanced," we refer to the fact that the branches of the dendrogram contain a relatively even number of observations, which facilitates a more coherent and interpretable clustering solution.

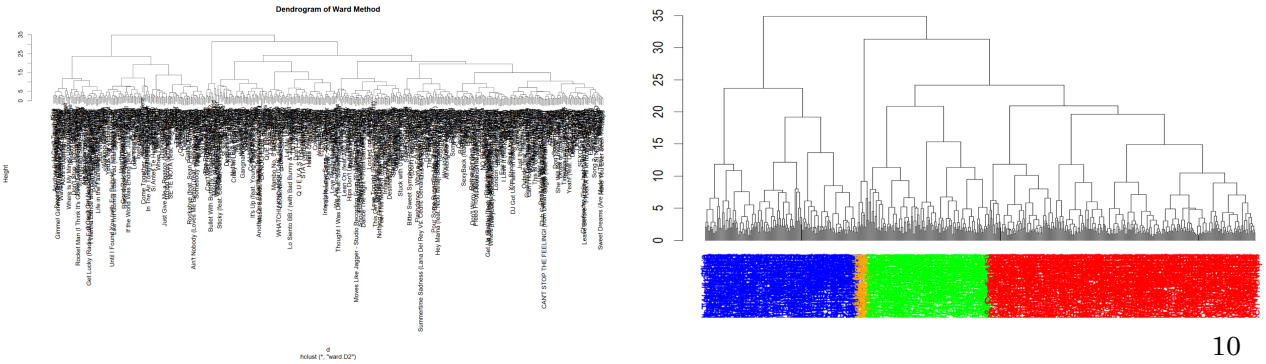

Figure 14: Ward Dendrogram *Figure 15: Ward Dendrogram Colored by cluster (k=4)*

As we can observe, the groups generated by the initial dendrogram are not particularly interpretable. Therefore, in order to gain a better intuition about the most appropriate number of clusters, we examined the clustering results for different values of k, ranging from 3 to 8. After analyzing these outcomes, we found that setting $k = 4$ appeared to provide the most meaningful separation between groups, although it is not a perfect solution. We present the resulting structure for the 4-cluster configuration, which offers the clearest segmentation given the data and dendrogram used.

To further confirm the optimal number of clusters, we conducted an evaluation using both the Pseudo-F Index and the Silhouette method. The analysis of the Silhouette values did not yield interpretable results (see Appendix), so we turned our attention to the Pseudo-F Index. In contrast, the Pseudo-F Index provided a more comprehensible and reliable indication of cluster structure, which supported our decision regarding the number of clusters.

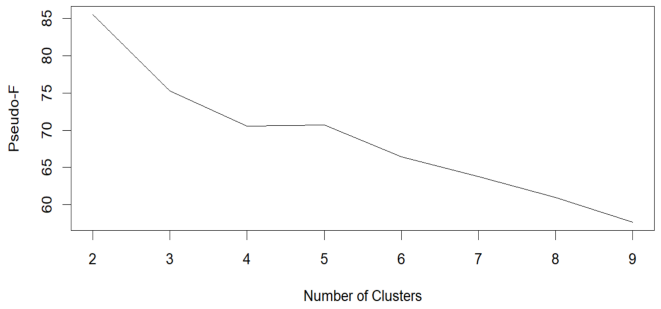

Figure 16: Pseudo-F Index

We can observe that the "elbow" of the Pseudo-F index curve occurs at $k = 4$, which confirms our previous assumption that the optimal number of clusters is four. Based on this finding, we proceeded to apply both the K-Means algorithm and Hierarchical Clustering on Principal Components (HCPC).

When applying K-Means with $k = 4$, we obtained the results shown in Appendix 8. However, upon analyzing the resulting clusters, we observed that there was no clear separation between them, making it difficult to identify meaningful distinctions or patterns. Due to the limitations in interpretability, we decided to explore an alternative clustering method by applying Hierarchical Clustering on Principal Components (HCPC). We now proceed to analyze the clusters produced by the HCPC approach, the results of which are also presented in Appendix 9.

The first difference we observe is that, instead of generating four clusters, which we had previously identified as the optimal number, HCPC produces only three. Despite this, we still see that the clusters are largely overlapping, making them, much like in the K-Means case, virtually uninterpretable.
Given this outcome, we decided to return to the initial clusters generated by the dendrogram for our analysis. To analyze the generated clusters, we will examine the group means for each cluster.

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4
energy	-0.9350006	0.4270083	0.1971109	0.4826896
tempo	-0.2855186	-0.1143830	0.5977074	0.1804946
danceability	-0.03201162	-0.03983386	0.13500664	- 0.070823
loudness	-0.9316033	0.4165706	0.3026000	-0.463589

liveness	-0.2944352	0.2067957	-0.0603825	-0.254620
valence	-0.11690891	0.08569555	-0.05877350	0.1921502
speechiness	-0.3871572	-0.3189819	1.2318660	-0.404166
track_popularity	0.0343136	0.1859125	-0.4166813	-0.407338
instrumentalness	-0.0915467	-0.1517466	-0.1356500	6.4005403
duration_ms	0.14397876	-0.03661306	-0.20585600	1.1382189
acousticness	0.8871125	-0.3894927	-0.1999842	-0.6934019

Table 2. *Group means by cluster*

Cluster 1: Calm and Acoustic
This group includes songs characterized by very low energy, slow tempo, low volume (loudness) and limited live presence (liveness). However, it stands out for its high level of acousticness, indicating that these are highly acoustic, natural, and relaxed tracks. Overall, this cluster is most closely associated with calm and acoustic. (Another Love)

Cluster 2: Popular and Powerful
This cluster consists of songs with high loudness, high energy, strong live presence (liveness), and notable popularity. As such, it represents a group that is commercially oriented and geared toward mainstream success. On the other hand, it shows very low levels of instrumentalness, suggesting that these tracks are primarily vocal, follow a traditional structure, and receive substantial media attention. (Sweet Child O'Mine)

Cluster 3: Rhythmic and Spoken
This group features songs with the highest tempo, high danceability, and strong vocal presence (speechiness), pointing to genres such as rap, spoken word, or other experimental formats. Despite their energetic and rhythmic nature, these songs tend to have low popularity, likely due to their affiliation with underground or alternative music scenes. Most of the other variables remain close to the average. (The Viper)

Cluster 4: Instrumental and Emotional
This final group stands out for its high energy, long duration, positive emotional tone (valence), and high instrumentalness. In contrast, these songs are less danceable, feature minimal vocals (low speechiness), and are the least popular. They also exhibit very low acousticness, indicating that they may be more electronically produced or designed as ambient/introspective pieces. (Around the World.

Discriminant Analysis (LDA & QDA) and MANOVA

Based on the description of an LDA & QDA, we begin by assessing the normality of our data using the Shapiro-Wilk test, Kolmogorov-Smirnov test, Q-Q plots, and histograms (Appendix 19 & 20). Both tests reject the null hypothesis, indicating a lack of normality across all variables. However, since p-values can be unreliable in large datasets, we proceed to examine the histograms and Q-Q plots for further confirmation.

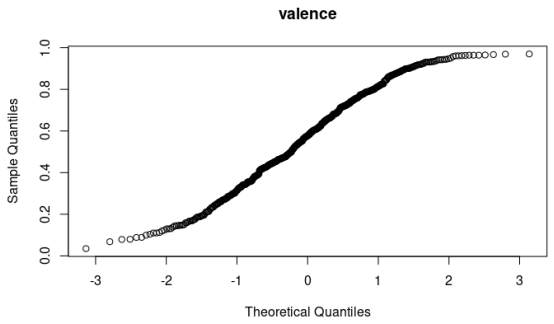

Figure 17: QQ plot of valence

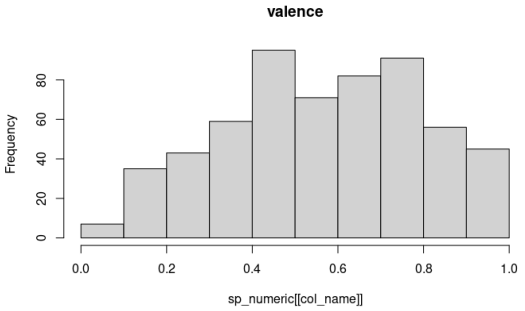

Figure 18; histogram of valence

After investigation, the variable *valence* is the one that most closely resembles a normal Q-Q plot; however, it still deviates from the diagonal line, indicating departures from normality. Additionally, its histogram does not exhibit the typical shape of a Gaussian distribution.
We perform Box's M test to assess whether the covariance matrices of the predictor variables are equal across the groups defined by the target variable *playlist_genre*.

```
        Box's M-test for Homogeneity of Covariance Matrices

data:  sp_numeric
Chi-Sq (approx.) = 856.05, df = 264, p-value < 2.2e-16
```

Figure 19: Box's M test results

A p-value near zero indicates that the covariance matrices differ significantly across groups, and this result is visually supported by the boxplots (Appendix 21). Since both LDA and QDA assume normality, a data transformation is necessary. However, because LDA also requires equal covariance matrices, which is not the case here, we will proceed with QDA.

After applying a Box-Cox transformation to all variables and re-evaluating normality using both tests, Q-Q plots, and histograms, we observe that the Shapiro-Wilk test continues to reject the null hypothesis for all variables, whereas the Kolmogorov-Smirnov test accepts normality for most of them. Upon inspecting the Q-Q plots and histograms, we identify the variable that deviates most clearly from a normal distribution, based on its visual pattern.

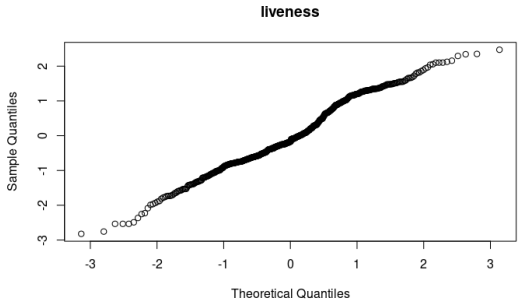

Figure 20: qq plot of liveness

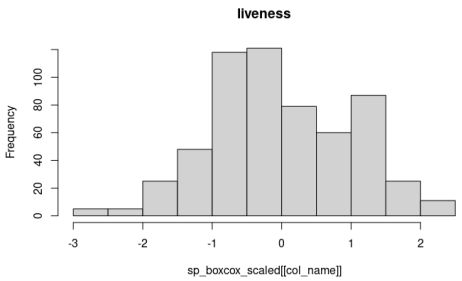

Figure 21; histogram of liveness

Therefore, it seems reasonable to conclude that our data can be considered approximately normal (Appendix 22), allowing us to proceed with the QDA analysis from this point forward.
We used MANOVA (Appendix 23) to assess the relative importance of the variables and found that *liveness* and *tempo* appear to be irrelevant, as indicated by their high p-values.

```
Response liveness :
                Df  Sum Sq  Mean Sq F value Pr(>F)
playlist_genre   4  0.0568 0.014208  0.8163 0.5151
Residuals      579 10.0779 0.017406
```

```
Response tempo :
                Df Sum Sq Mean Sq F value Pr(>F)
playlist_genre   4   1675  418.78  0.5911 0.6692
Residuals      579 410230  708.52
```

Table 3: MANOVA results for liveness and tempo

Although we attempted to improve the QDA and Naive Bayes models by removing these variables, the best performance was still achieved when using all variables. Nevertheless, this insight proved useful for fitting the Stepwise Classification model.

From the QDA results, we obtain the means of each variable across the five groups, allowing us to analyze their discriminatory power. A variable is considered to make a significant contribution when the group means differ substantially from one another.

```
Group means:
                energy       tempo danceability     liveness     valence speechiness track_popularity duration_ms acousticness
electronic  0.31484653  0.16088331  -0.05049866  0.108325803 -0.37405815  -0.1084372      -0.08768052 -0.06470154  -0.10976184
hip-hop    -0.34412672 -0.02562810   0.56552651  0.283271417 -0.77224653   1.0580517      -0.14676849 -1.02168902  -0.35307895
latin      -0.15070348 -0.03777783   0.58226458 -0.084894666  0.14483110   0.5284634      -0.13677042 -0.28423114   0.55989479
pop         0.02679741 -0.04243887  -0.05218998 -0.005880688  0.08167305  -0.2879545       0.21852485  0.10696967  -0.03918595
rock        0.09569330  0.07629407  -0.93453481 -0.038584360  0.11392729  -0.3984978      -0.25050483  0.59813525  -0.52993819
```

Table 4: QDA means across groups and variables

We observe that variables such as liveness, *tempo*, and *valence* have three groups with very similar means, suggesting they offer limited discriminative power for distinguishing between genres. In contrast, variables like *speechiness*, *duration_ms*, and *acousticness* show separated group means, indicating they are more effective predictors for genre classification. Creating the confusion table, we get the following results.

```
           electronic hip-hop latin pop rock
electronic         13       1     3  14    0
hip-hop             1      30     2   3    0
latin               8       5   111  22    4
pop                38       2    25 193   27
rock                0       0     2  16   64
[1] 0.7037671
```

Table 5: Confusion matrix for QDA and accuracy

The model classifies the data into the correct categories with high accuracy, as reflected by the strong diagonal dominance in the confusion matrix. The main exception lies in the confusion between *electronic* and *pop*, where 14 electronic tracks are misclassified as pop and 38 pop tracks as electronic. Despite this, the overall accuracy reaches **70.38%**, which is a solid result for a multi-class classification task.

Using the formula $Q = \frac{(n-n_y \cdot k)^2}{n \cdot (k-1)}$ where n is the total observations, n_y is the correct classifications, and k is the number of genres, we compute the Q statistic:

```
Q statistic: 926.3018
Critical value (chi-squared, 0.95, df = 4 ): 9.487729
```

Figure 22: Q test values

This statistic indicates whether the separation between classes is statistically significant. In our case, the computed Q value is much greater than the critical value, leading us to conclude that the classification model achieves a strong and meaningful separation between genres. Since stepwise classification typically adds or removes variables only when doing so leads to at least a 5% change in accuracy, and no variable in the model meets that threshold, the method proves ineffective when applied, whether using forward or backward selection. However, by leveraging the insights from the MANOVA results and excluding less relevant variables, the backward method consistently achieves an accuracy of 0.635, which is an improvement over the 0.62 obtained when using all variables.

Lastly, let's apply Naive Bayes, a probabilistic classification algorithm based on Bayes' theorem. After fitting the model to our data, we obtain the following confusion matrix, which reflects how well the algorithm performs in predicting genre categories across the dataset.

```
sp_naive.class electronic hip-hop latin pop rock
    electronic          8       0     1   9    0
    hip-hop             0      20     4   5    0
    latin               8      11   100  27    4
    pop                41       7    36 186   36
    rock                3       0     2  21   55
[1] 0.6318493
```

Table 6: Confusion matrix for Naive Bayes and accuracy

The Naive Bayes model performs worse than QDA, showing similar misclassification patterns between *electronic* and *pop*, and an increased number of errors between *pop* and *rock*. Overall, its classification accuracy is lower than that of QDA but comparable to the Stepwise Classification model, yielding a similar overall accuracy despite the different methodology.

Conclusions

Across the various multivariate techniques applied to the Spotify dataset, a coherent picture emerges in which three fundamental latent factors—energy versus acousticness, danceability versus instrumental complexity, and speechiness versus mainstream popularity—consistently distinguish musical genres. Principal Component Analysis reduces ten numerical audio features to four interpretable axes that separate high-energy, loud, and commercial tracks from more acoustic, mellow recordings; contrast danceable, concise songs with longer, instrumentally rich pieces; and juxtapose lyric-driven, less mainstream tracks against widely popular melodic ones.

Multidimensional Scaling corroborates this structure in two dimensions, revealing a "rhythmic/expressive intensity" axis (danceability and valence) and a "lyrical density versus mainstream appeal" axis (speechiness and popularity), with categorical genre information via Gower distance further clarifying separations among Pop, Latin, Rock, and Hip-hop.

Multiple Correspondence Analysis, by binning numerical features into categories, again highlights high danceability/speechiness against high instrumentalness/acousticness and contrasts high energy/valence with lower energy/valence, reproducing the same genre clusters.

Hierarchical clustering identifies four coherent groups—acoustic/calm, popular/powerful, rhythmic/rap, and instrumental/emotional—that align directly with these latent dimensions.

Finally, Quadratic Discriminant Analysis achieves approximately 70 percent accuracy in classifying songs by genre, confirming that speechiness, acousticness, and track duration are the most discriminative variables, while liveness and tempo contribute minimally.

A practical limitation we encountered was the **large number of songs and genres**, which made some visualizations overly dense and hard to interpret. To address this, we **restricted our analysis to five major genres**, ensuring clearer patterns and more meaningful comparisons across methods.

In sum, these convergent findings reveal that popular Spotify tracks are organized along a tripartite continuum of **sonic intensity, rhythmic accessibility, and lyrical focus**, which neatly partitions genres into recognizable and interpretable segments.

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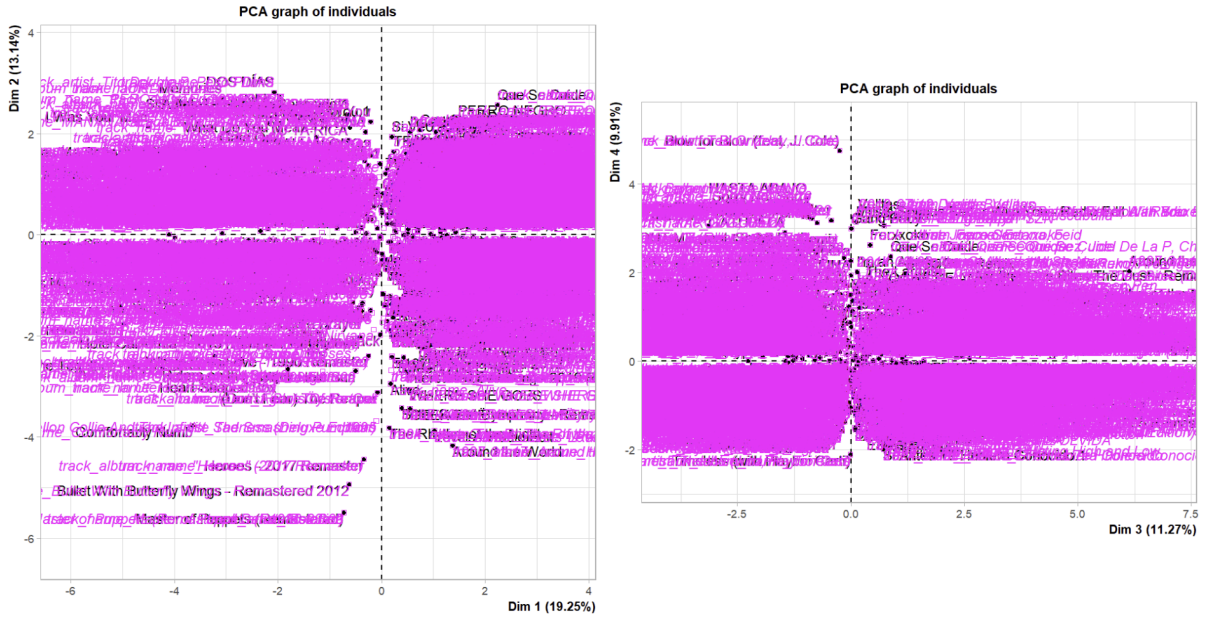
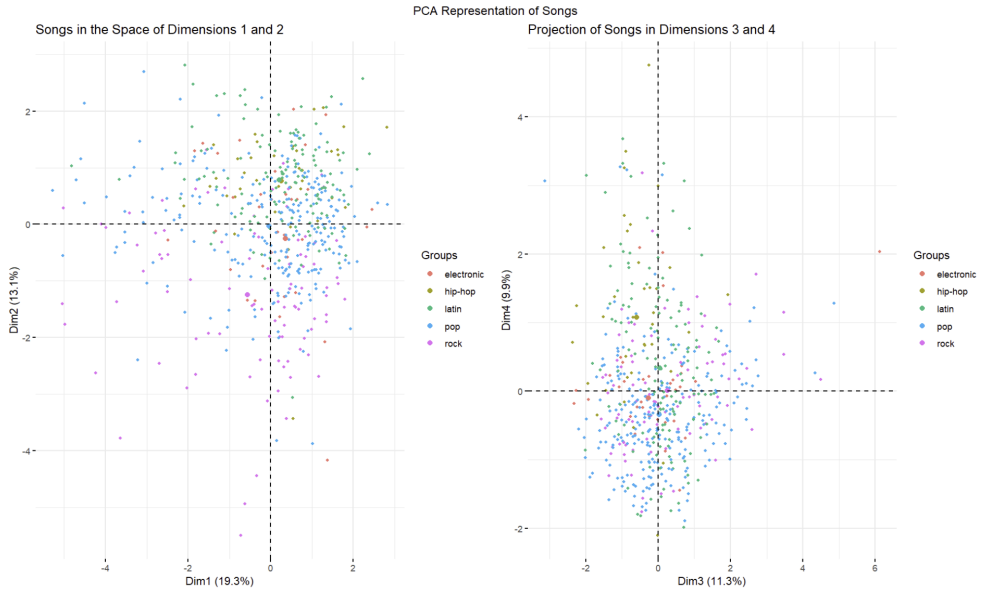
Appendix

PCA



Annex 1: PCA graph of individuals with labels



Appendix 2 PCA representation of songs by genre

Genre	Dim1 (Mean ± SD)	Dim2 (Mean ± SD)	Dim3 (Mean ± SD)	Dim4 (Mean ± SD)	n
Electronic	0.35 ± 0.96	-0.25 ± 1.05	-0.27 ± 1.22	-0.10 ± 0.77	60
Hip-hop	0.26 ± 1.10	0.77 ± 0.91	-0.59 ± 0.83	1.07 ± 1.33	38
Latin	0.21 ± 1.18	0.79 ± 0.85	0.05 ± 0.81	0.34 ± 1.23	143

Pop	-0.03 ± 1.58	-0.03 ± 0.96	0.03 ± 1.17	-0.34 ± 0.82	248
Rock	-0.56 ± 1.73	-1.24 ± 1.29	0.26 ± 1.29	0.01 ± 0.83	95

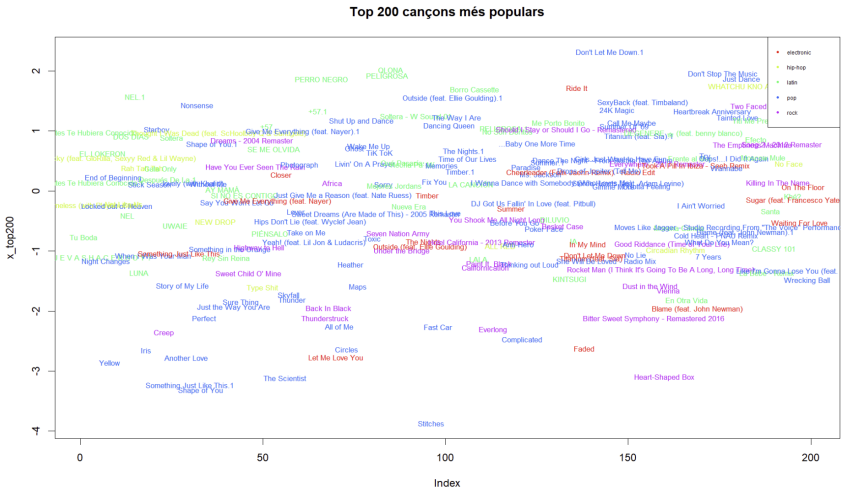
Appendix 3: Genre Distribution Across PCA Dimensions

The distribution of genres across the PCA dimensions supports previous interpretations. On **Dimension 1**, associated with sonic intensity and commercial potential, *rock* exhibits the lowest mean, suggesting a less mainstream and more acoustic profile, while *electronic* and *hip-hop* lead with higher scores. **Dimension 2**, interpreted as an axis of *danceability versus structural complexity*, clearly separates *rock* (low values) from the more rhythmically accessible *latin* and *hip-hop* tracks (high values). On **Dimension 3**, which captures *musical expressiveness versus commercial simplicity*, *rock* and *latin* show higher means, aligning with more complex musical content, while *hip-hop* scores lowest, indicating more formulaic, production-driven structures. **Dimension 4** reflects the *lyrical versus melodic* axis. Here, *hip-hop* and *latin* dominate with high scores due to their strong vocal and rhythmic content, whereas *pop* and *electronic* lean toward melodic simplicity and lower speech presence.

Standard deviations reveal genre diversity: *pop* and *rock* exhibit greater internal variability across all dimensions, suggesting stylistic breadth. Conversely, *hip-hop* and *latin* are more cohesive, especially in Dimensions 2 and 3, reinforcing their consistent rhythmic and expressive traits.

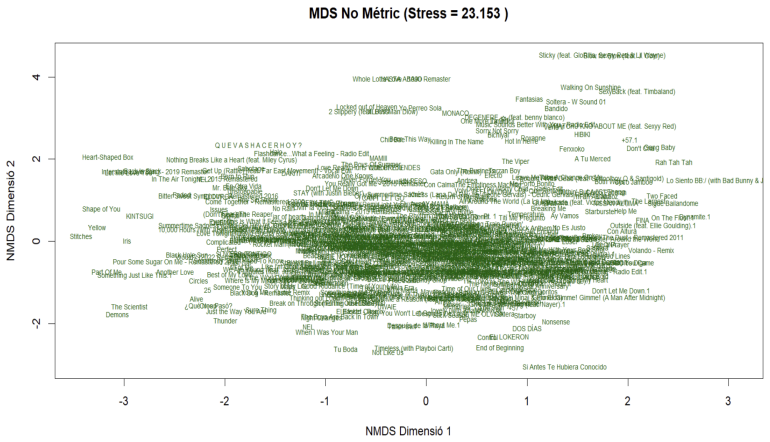
In sum, *hip-hop* emerges as lyrically dense and rhythmically driven, *rock* as less commercial and more structurally complex, while *pop* and *electronic* balance commercial appeal with stylistic diversity.

MDS



Appendix 4: MDS of the top 200 most popular songs

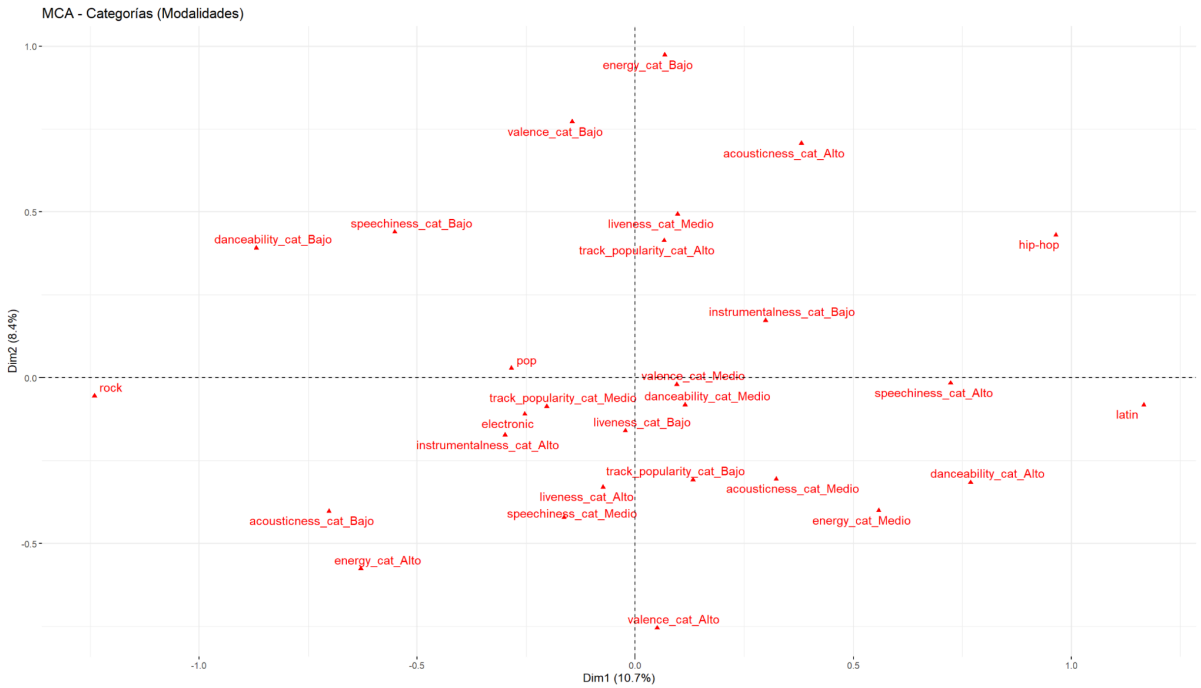
Non metric MDS



Appendix 5: Non-metric MDS

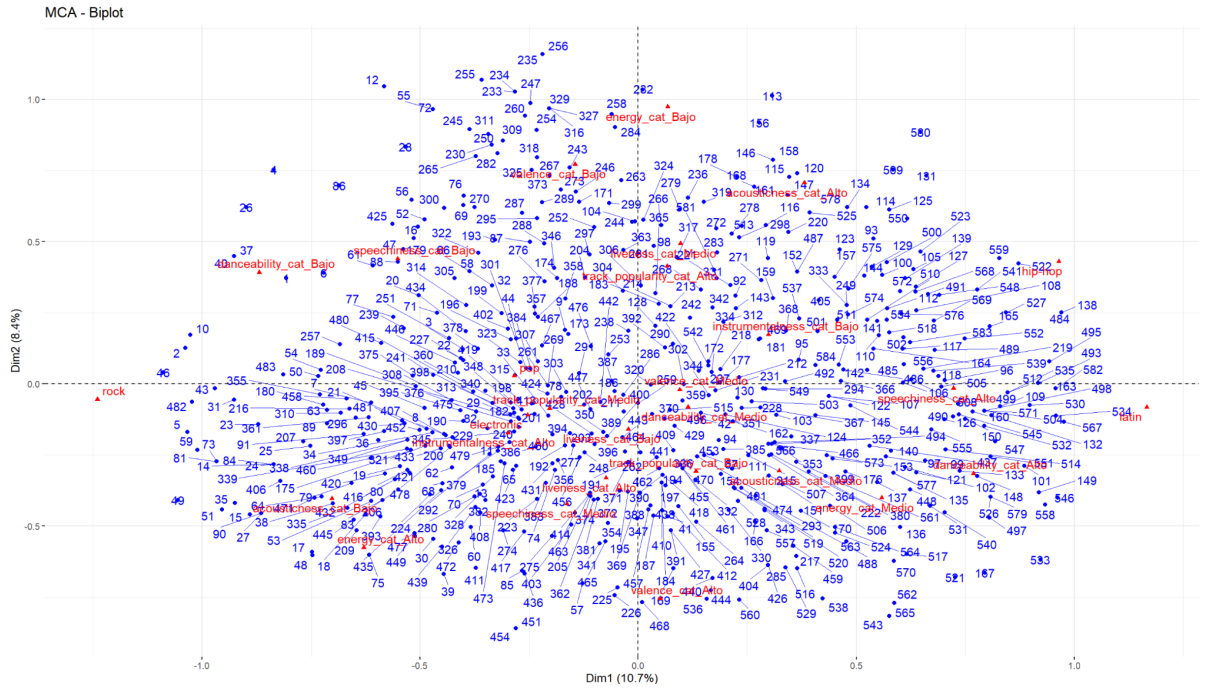
The non-metric multidimensional scaling (NMDS) plot provides a two-dimensional representation of the dissimilarities between songs based on their musical features. With a stress value of 23.153, the projection reflects a moderate distortion from the original high-dimensional space, indicating that some caution is needed when interpreting spatial proximities. Despite this, certain structural patterns can be observed. The central cluster denotes a concentration of songs with relatively similar characteristics, while more isolated songs on the periphery exhibit greater uniqueness or divergence from the main corpus. The presence of tight clusters in certain regions suggests that groups of songs share common acoustic or compositional traits. Additionally, the dispersion across both NMDS dimensions highlights the diversity present within the dataset, pointing to a wide range of stylistic elements. Overall, the NMDS visualization reveals meaningful, albeit approximate, relationships among songs and offers a useful perspective on the structure of musical similarity in the dataset

MCA



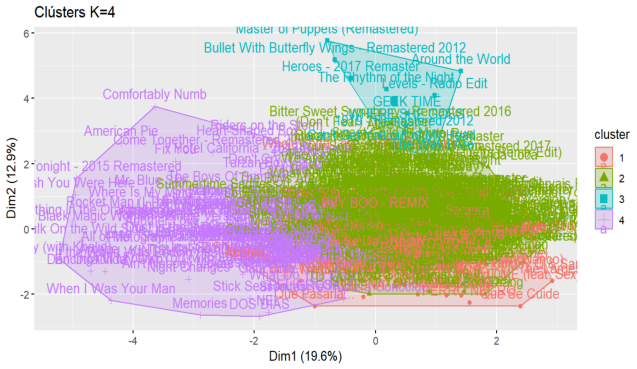
Appendix 6: MCA-Categories

As we can see we do not detect a U form, that means that we can discard the Guttman Effect.

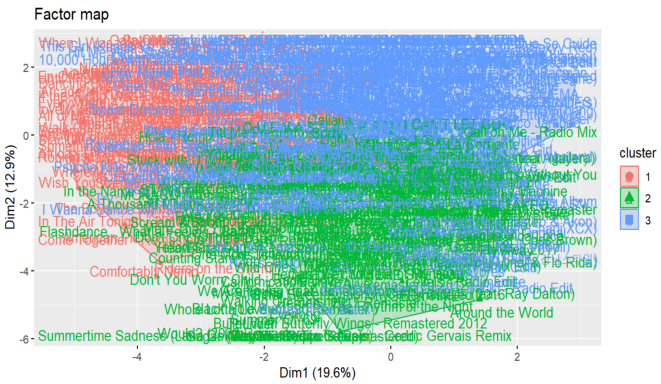


Appendix 7: MCA-Plot

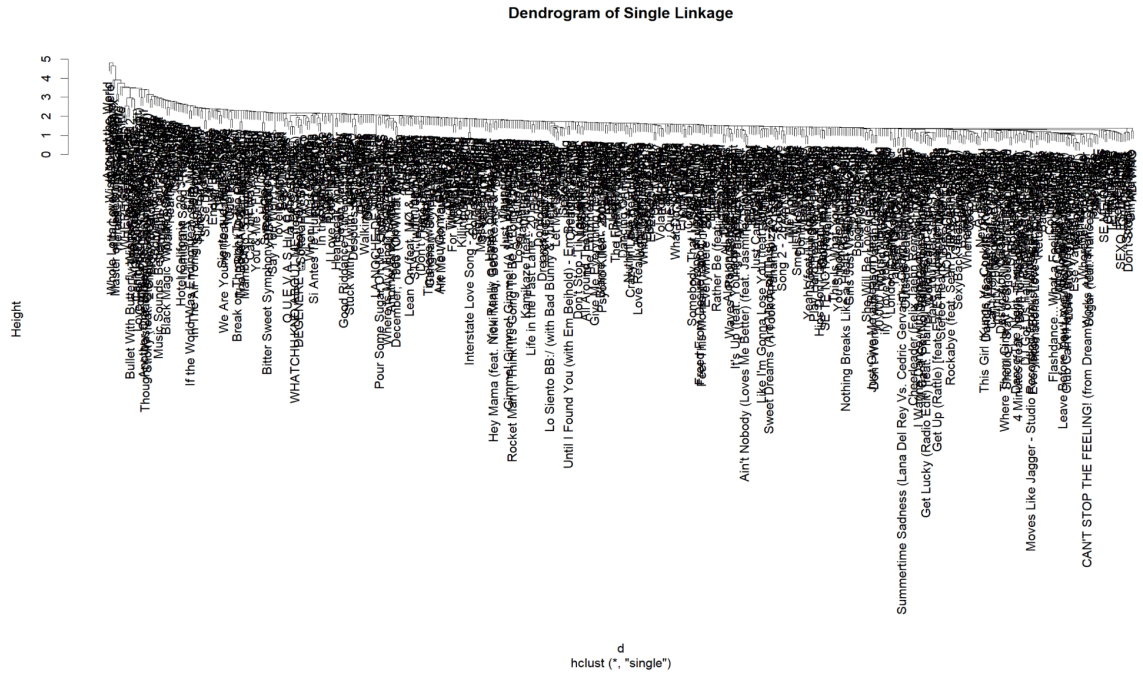
Cluster Analysis



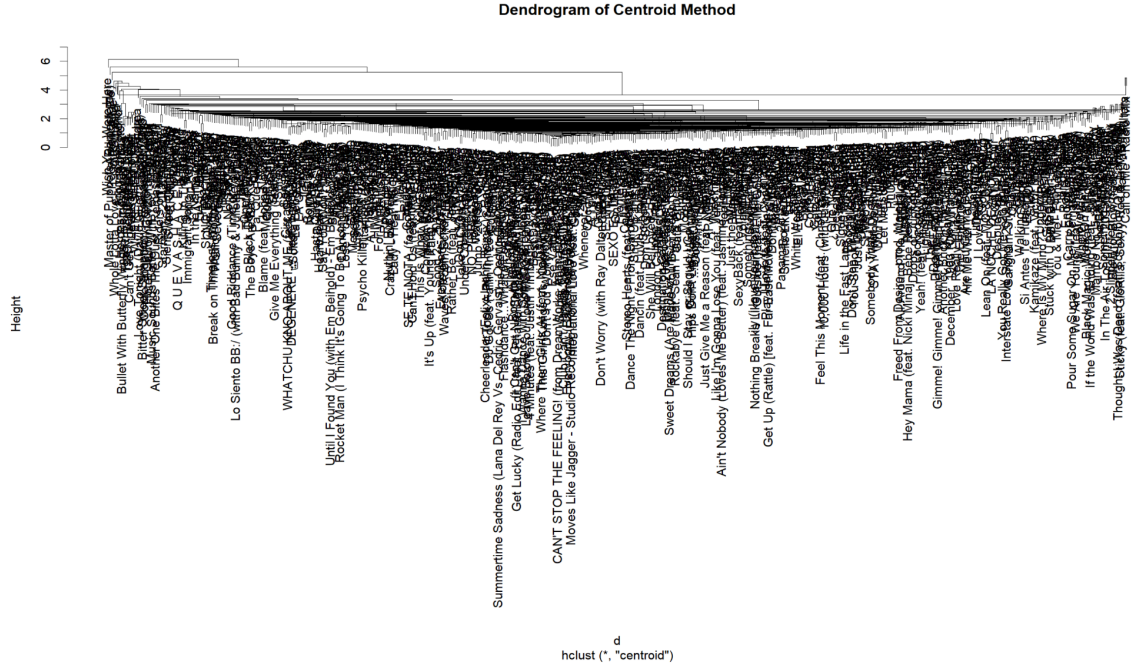
Appendix 8: K-Means Clusters



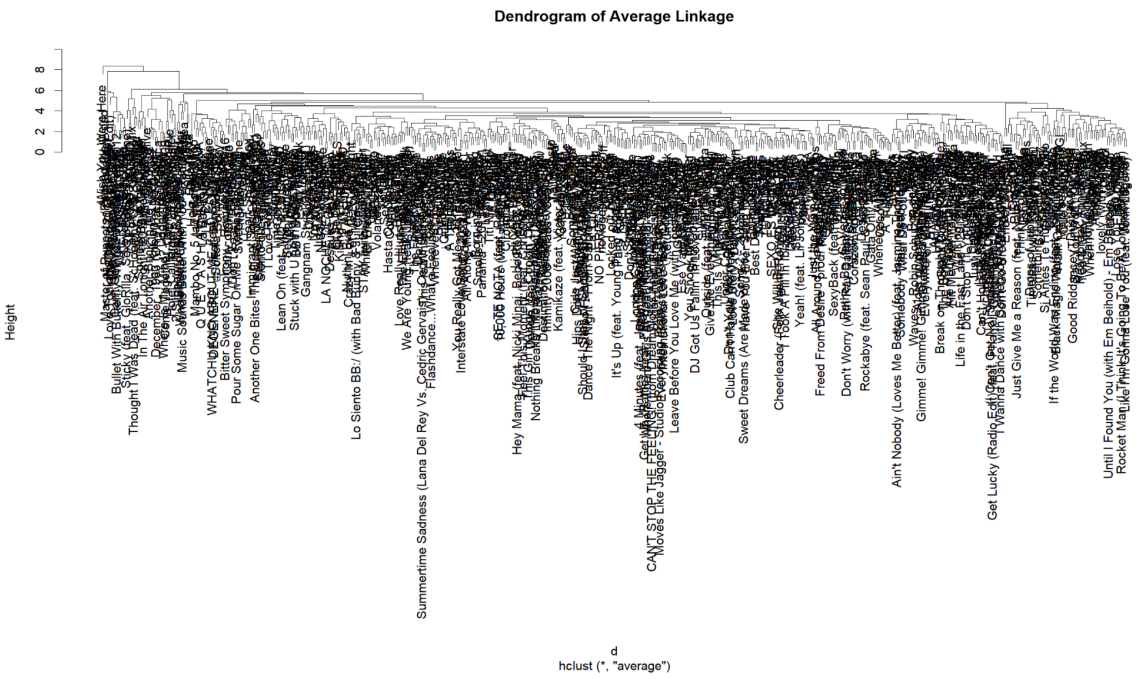
Appendix 9: HCPC Clusters



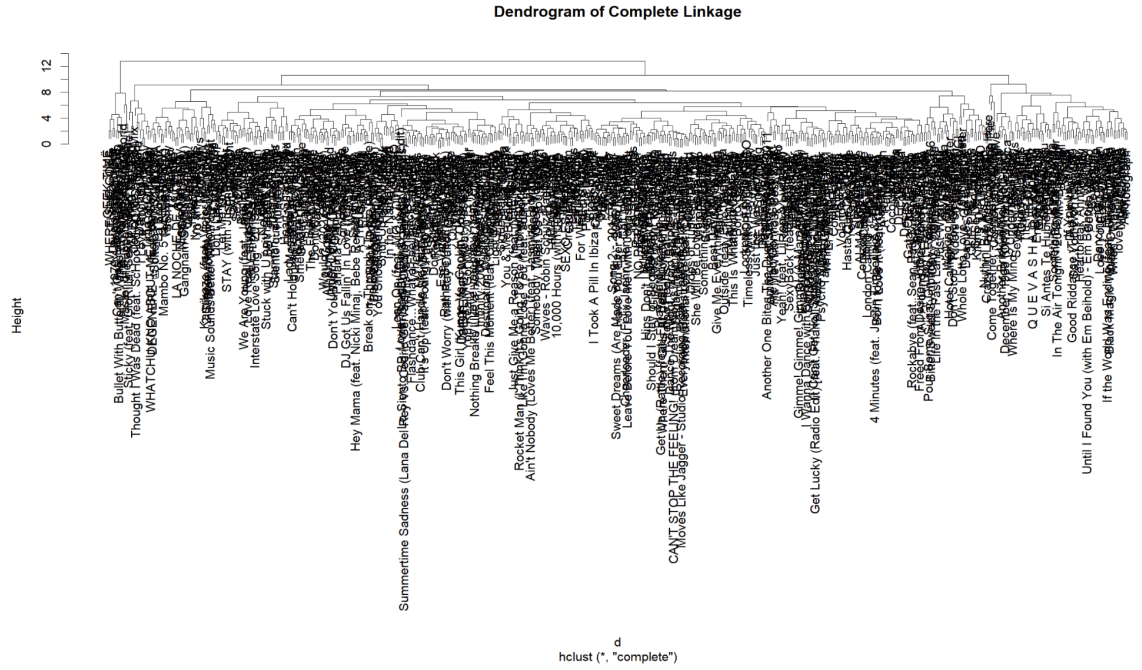
Appendix 10: Single Linkage Dendrogram



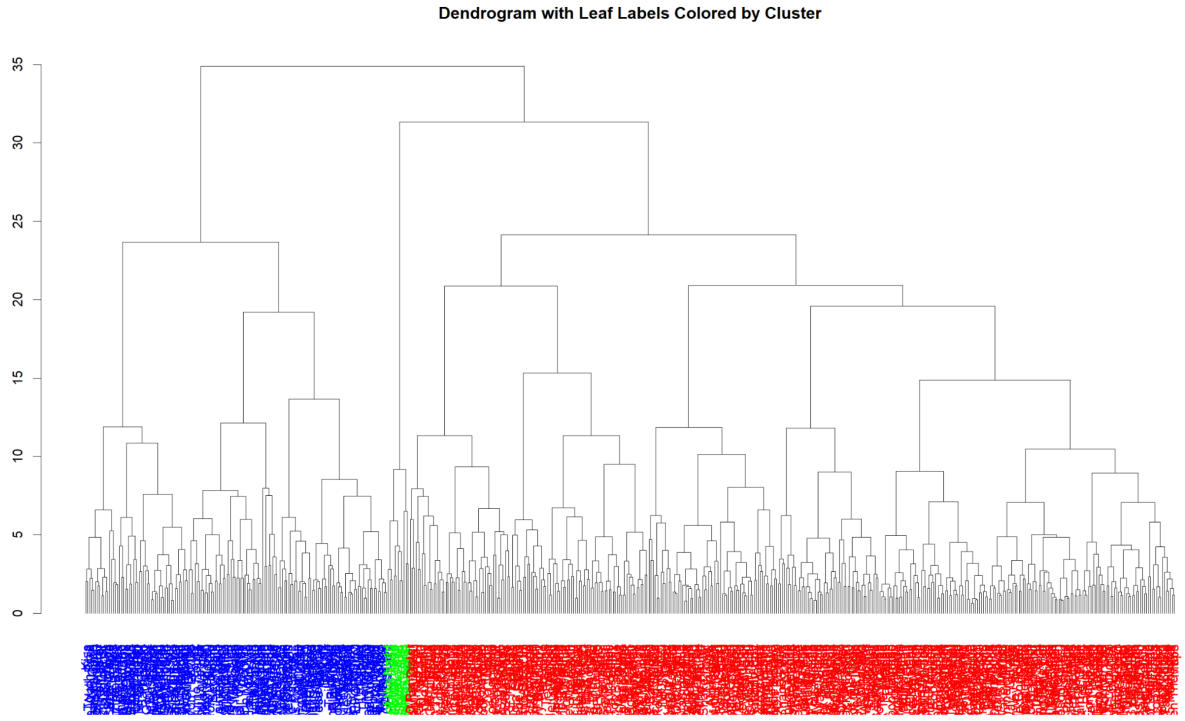
Appendix 11: Centroid Method Dendrogram



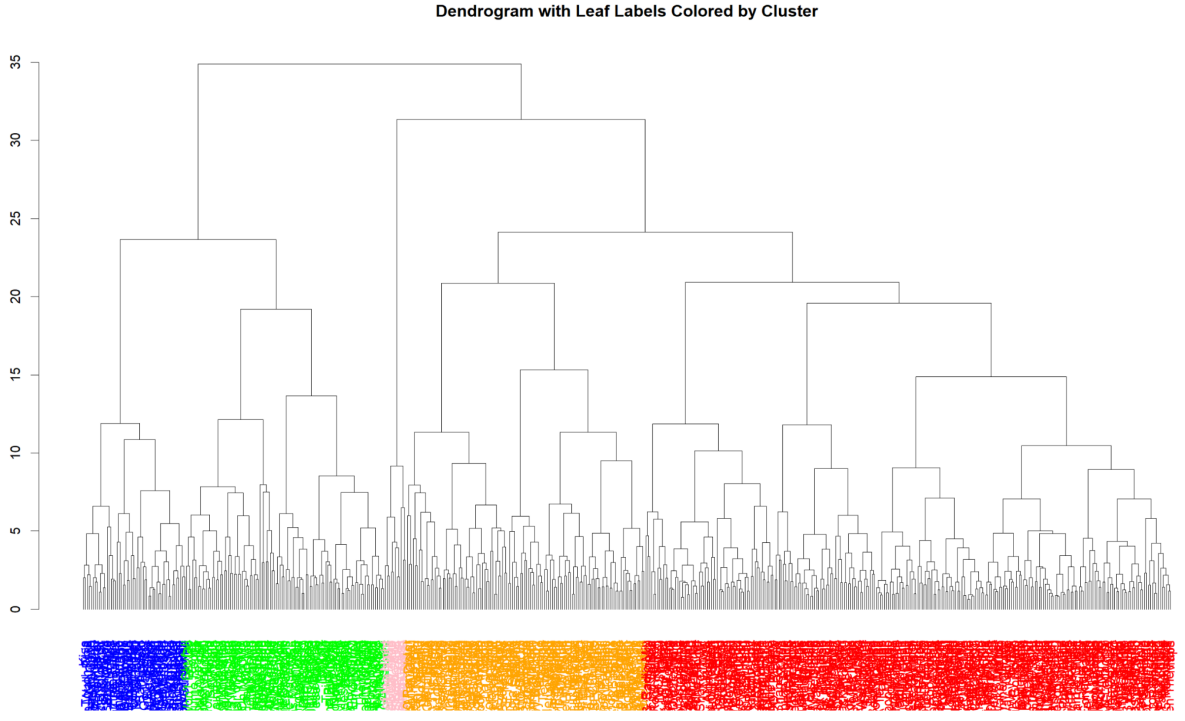
Appendix 12: Average Linkage Dendrogram



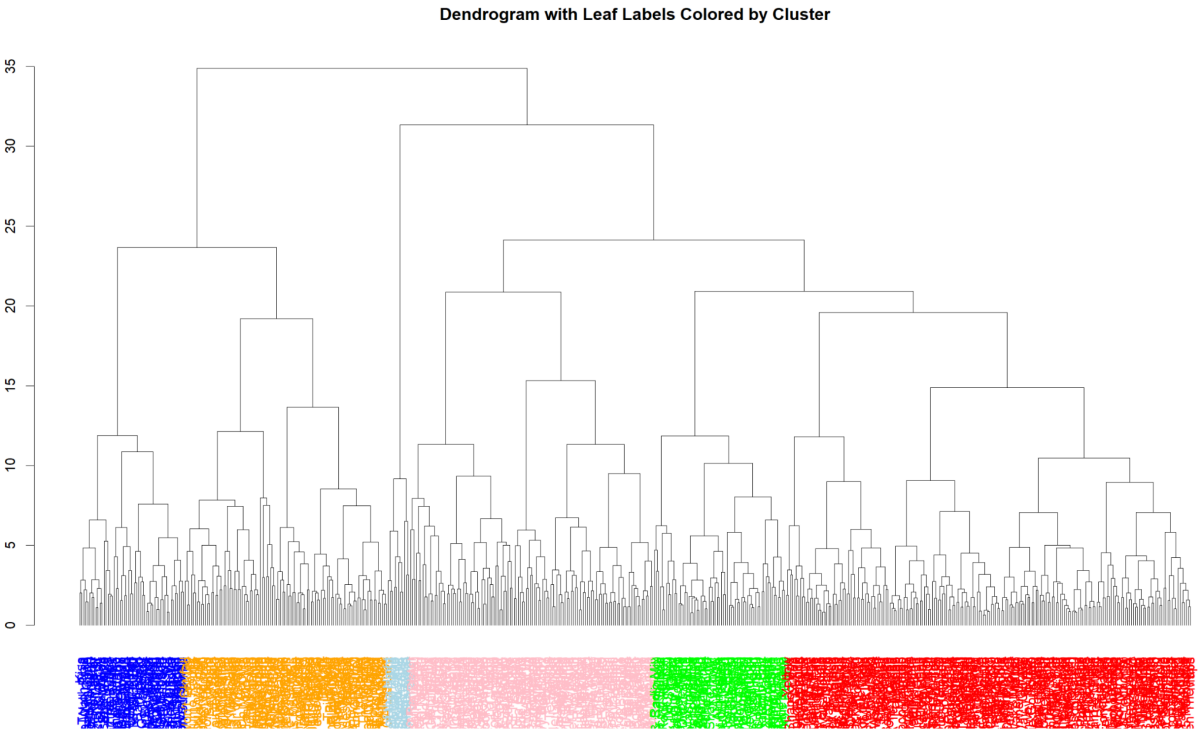
Appendix 13: Complete Linkage Dendrogram



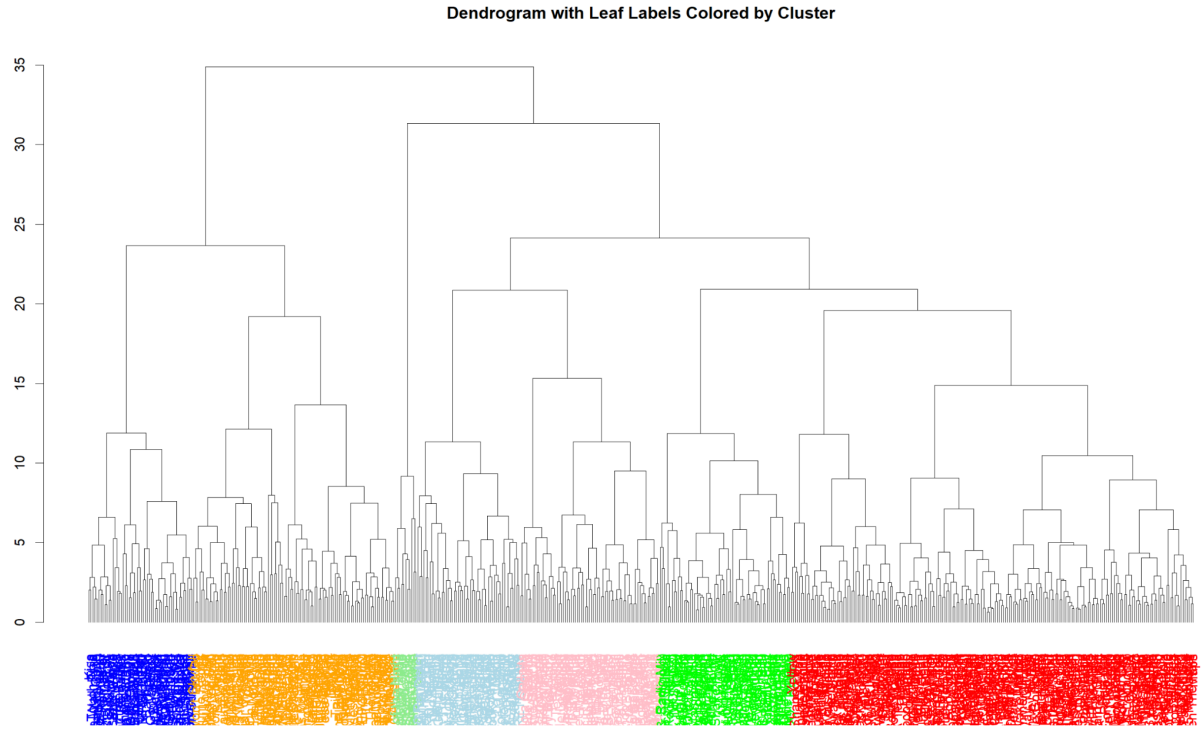
Appendix 14: 3-colored Ward dendrogram



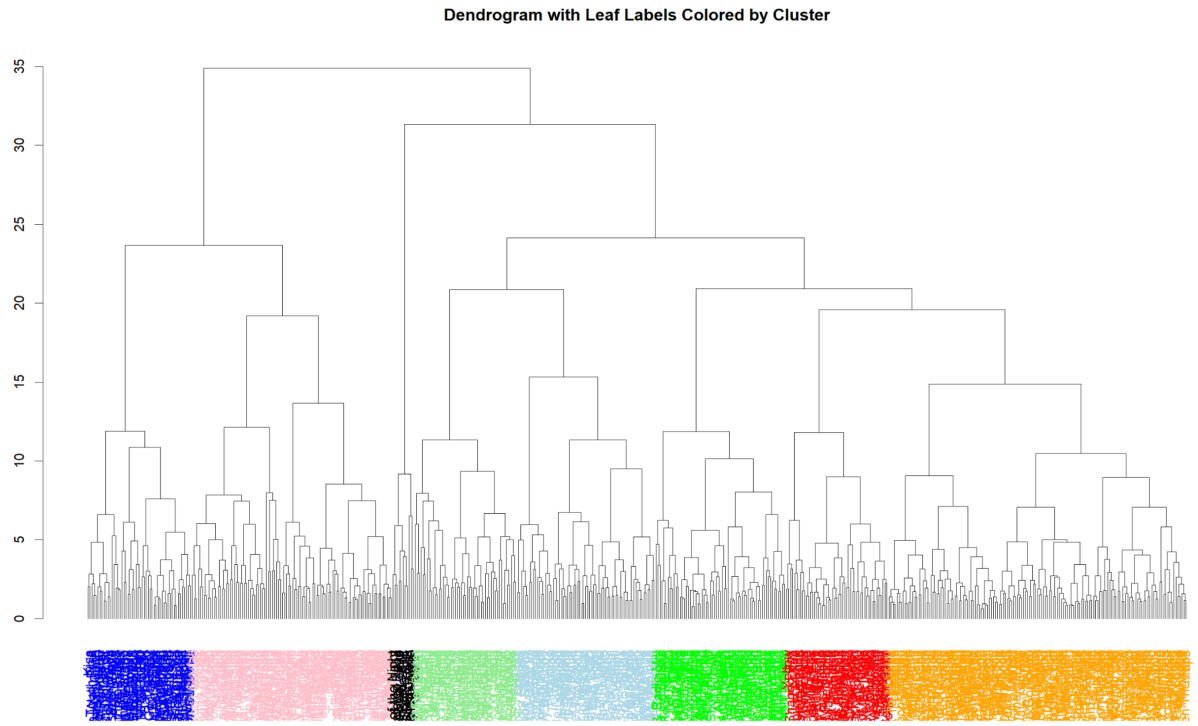
Appendix 15: 5-colored Ward dendrogram



Appendix 16: 6-colored Ward dendrogram



Appendix 17: 7-colored Ward dendrogram

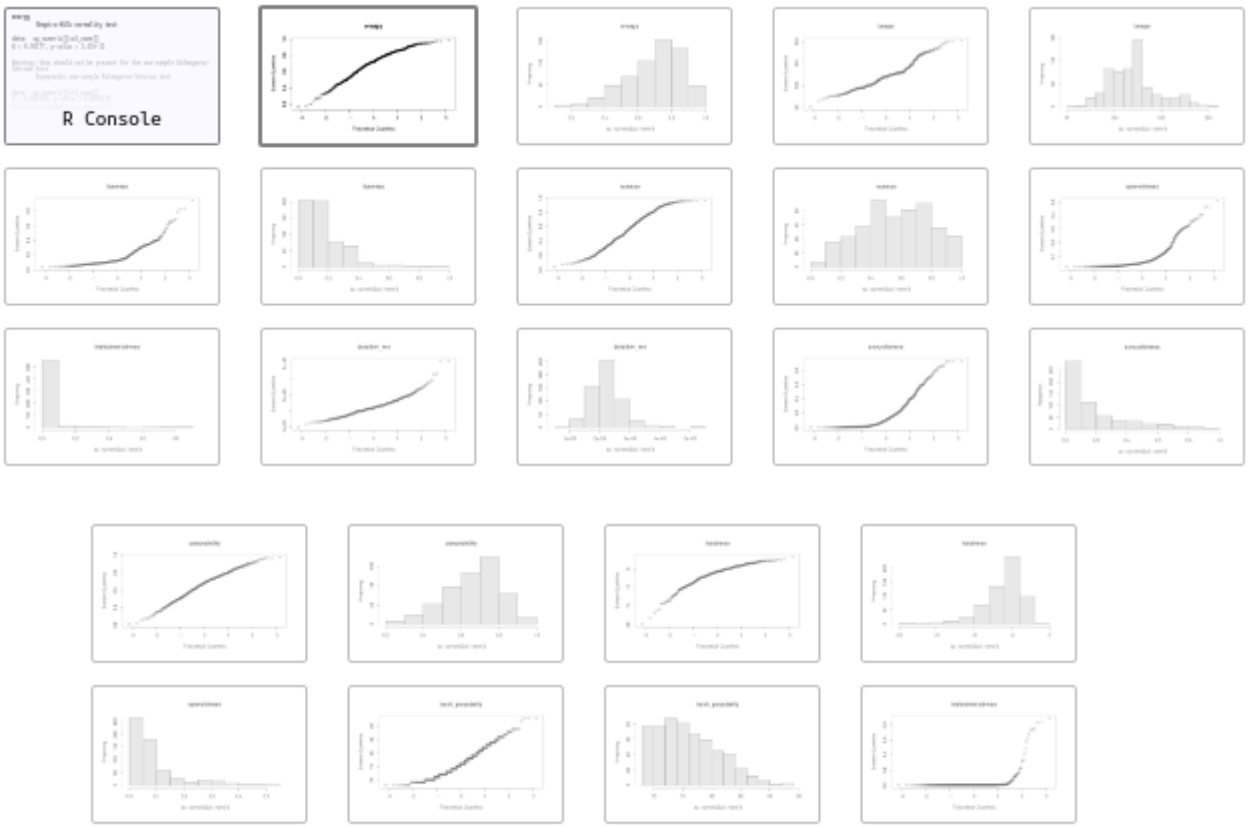


Appendix 18: 8-colored Ward dendrogram

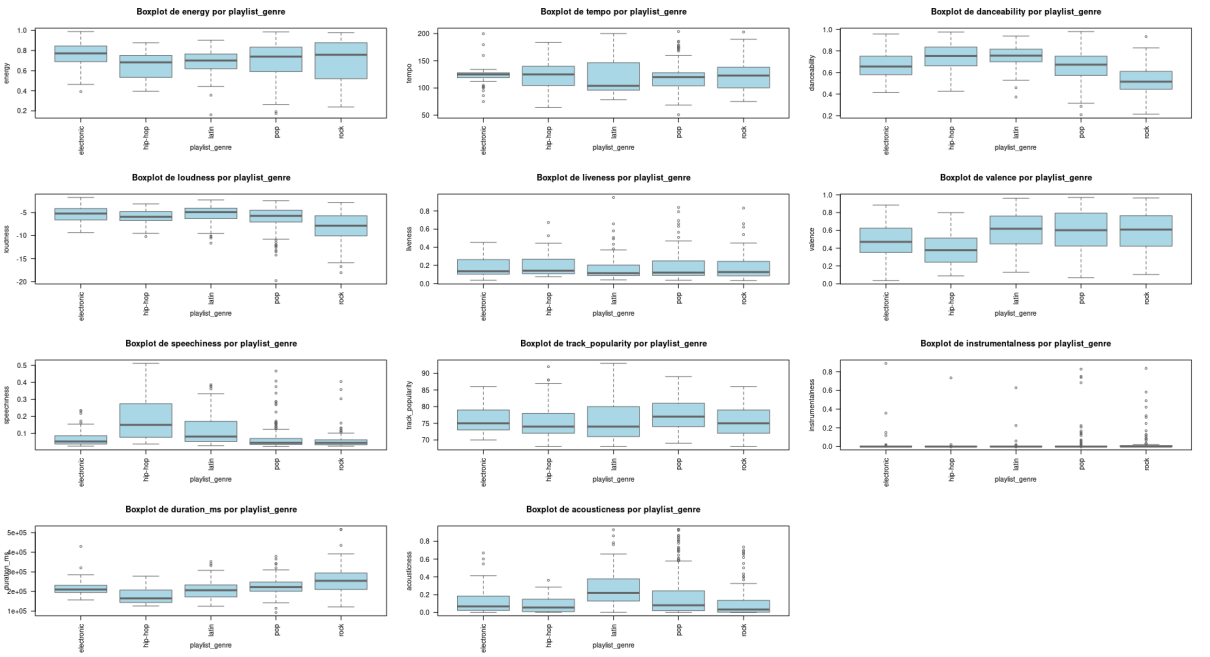
QDA & MANOVA

```
energy
        Shapiro-Wilk normality test

data:  sp_numeric[[col_name]]
W = 0.96177, p-value = 3.47e-11

Warning: ties should not be present for the one-sample Kolmogorov-Smirnov test
        Asymptotic one-sample Kolmogorov-Smirnov test

data:  sp_numeric[[col_name]]
D = 0.081162, p-value = 0.0009111
alternative hypothesis: two-sided

tempo
        Shapiro-Wilk normality test

data:  sp_numeric[[col_name]]
W = 0.95881, p-value = 1.026e-11

Warning: ties should not be present for the one-sample Kolmogorov-Smirnov test
        Asymptotic one-sample Kolmogorov-Smirnov test

data:  sp_numeric[[col_name]]
D = 0.12813, p-value = 9.4e-09
alternative hypothesis: two-sided
```

Appendix 19: Representative p-values of Shapiro-Wilk and Kolmogorov-Smirnov tests



Appendix 20: QQ-plots and histograms pre-box cox



Appendix 21: variance boxplots

Appendix 22: QQ-plots and histograms post-box cox

```
 Response energy :
               Df  Sum Sq  Mean Sq F value  Pr(>F)
playlist_genre   4  0.2722 0.068061  2.7256 0.02867 *
Residuals      579 14.4582 0.024971
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response tempo :
               Df Sq Mean Sq F value Pr(>F)
playlist_genre   4   1675  418.78  0.5911 0.6692
Residuals      579 410230  708.52

 Response danceability :
               Df Sq Mean Sq F value    Pr(>F)
playlist_genre   4 3.1779 0.79448  49.664 < 2.2e-16 ***
Residuals      579 9.2623 0.01600
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response loudness :
               Df Sq Mean Sq F value    Pr(>F)
playlist_genre   4  600.6  150.15  27.248 < 2.2e-16 ***
Residuals      579 3190.5    5.51
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response liveness :
               Df  Sum Sq  Mean Sq F value Pr(>F)
playlist_genre   4  0.0568 0.014208  0.8163 0.5151
Residuals      579 10.0779 0.017406

 Response valence :
               Df  Sum Sq Mean Sq F value    Pr(>F)
playlist_genre   4  1.9383 0.48458  9.7912 1.141e-07 ***
Residuals      579 28.6557 0.04949
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response speechiness :
               Df Sum Sq Mean Sq F value    Pr(>F)
playlist_genre   4 0.7508 0.18770  32.756 < 2.2e-16 ***
Residuals      579 3.3178 0.00573
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


 Response track_popularity :
               Df  Sum Sq Mean Sq F value   Pr(>F)
playlist_genre   4   493.1 123.264  4.6295 0.001098 **
Residuals      579 15416.2  26.626
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response instrumentalness :
               Df Sq  Mean Sq F value  Pr(>F)
playlist_genre   4 0.0900 0.022497  2.2088 0.06674 .
Residuals      579 5.8972 0.010185
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response duration_ms :
               Df     Sum Sq    Mean Sq F value    Pr(>F)
playlist_genre   4 2.2861e+11 5.7153e+10  24.588 < 2.2e-16 ***
Residuals      579 1.3458e+12 2.3244e+09
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

 Response acousticness :
               Df  Sum Sq Mean Sq F value    Pr(>F)
playlist_genre   4  1.6172 0.40430  9.7766 1.171e-07 ***
Residuals      579 23.9436 0.04135
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Appendix 23: MANOVA results